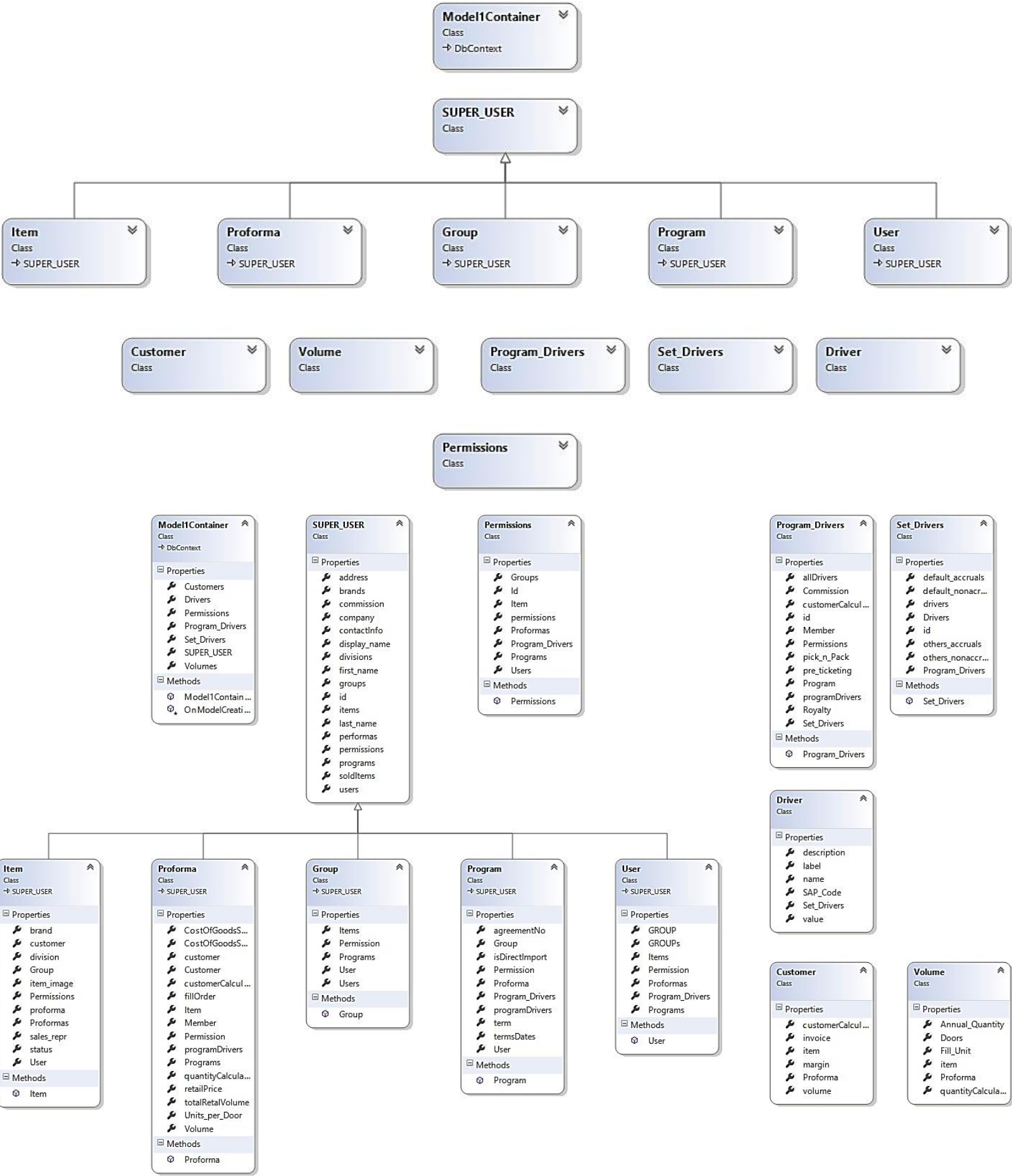
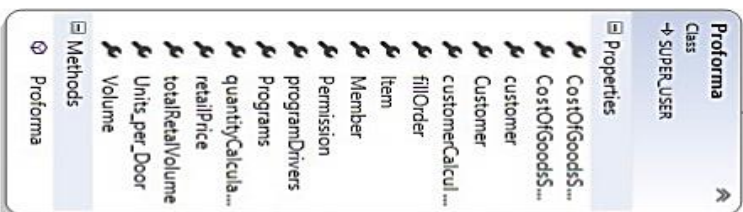
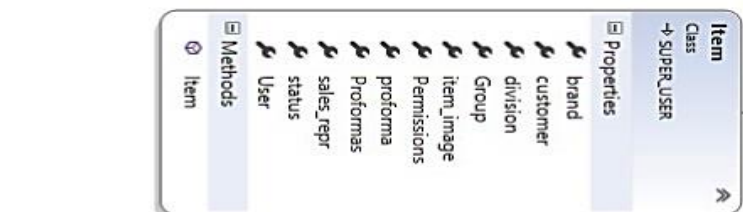
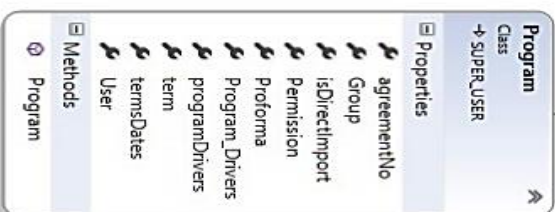
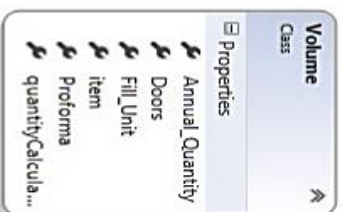
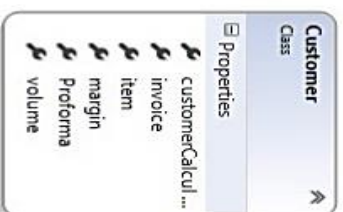
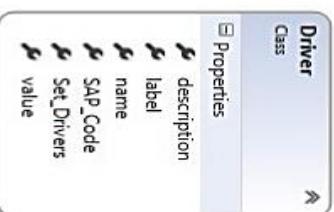
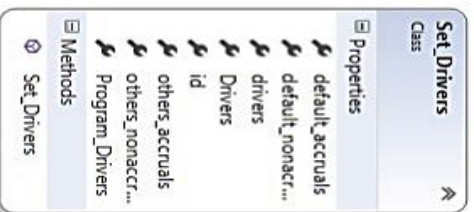
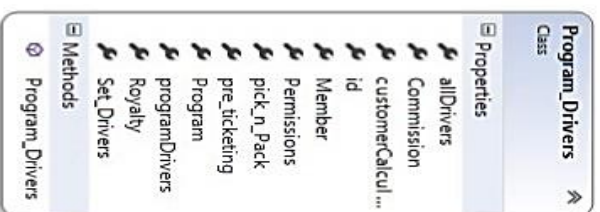
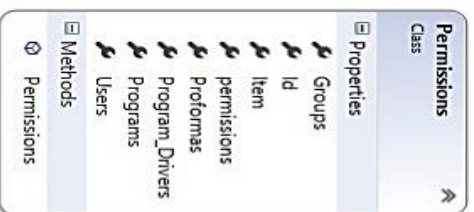
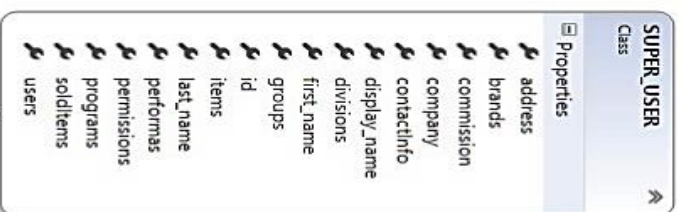


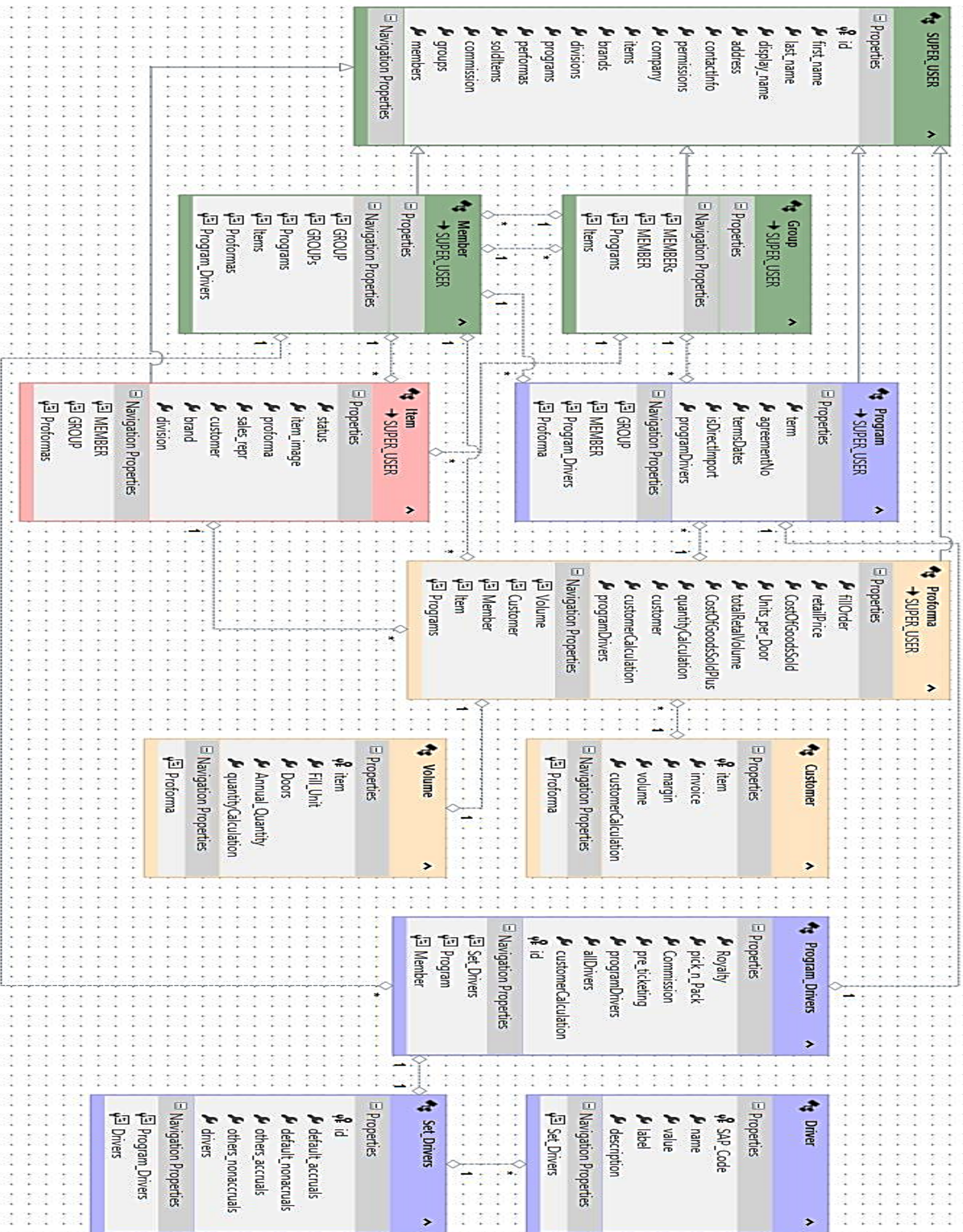
# LTB - Sales Portal

## Database Diagram

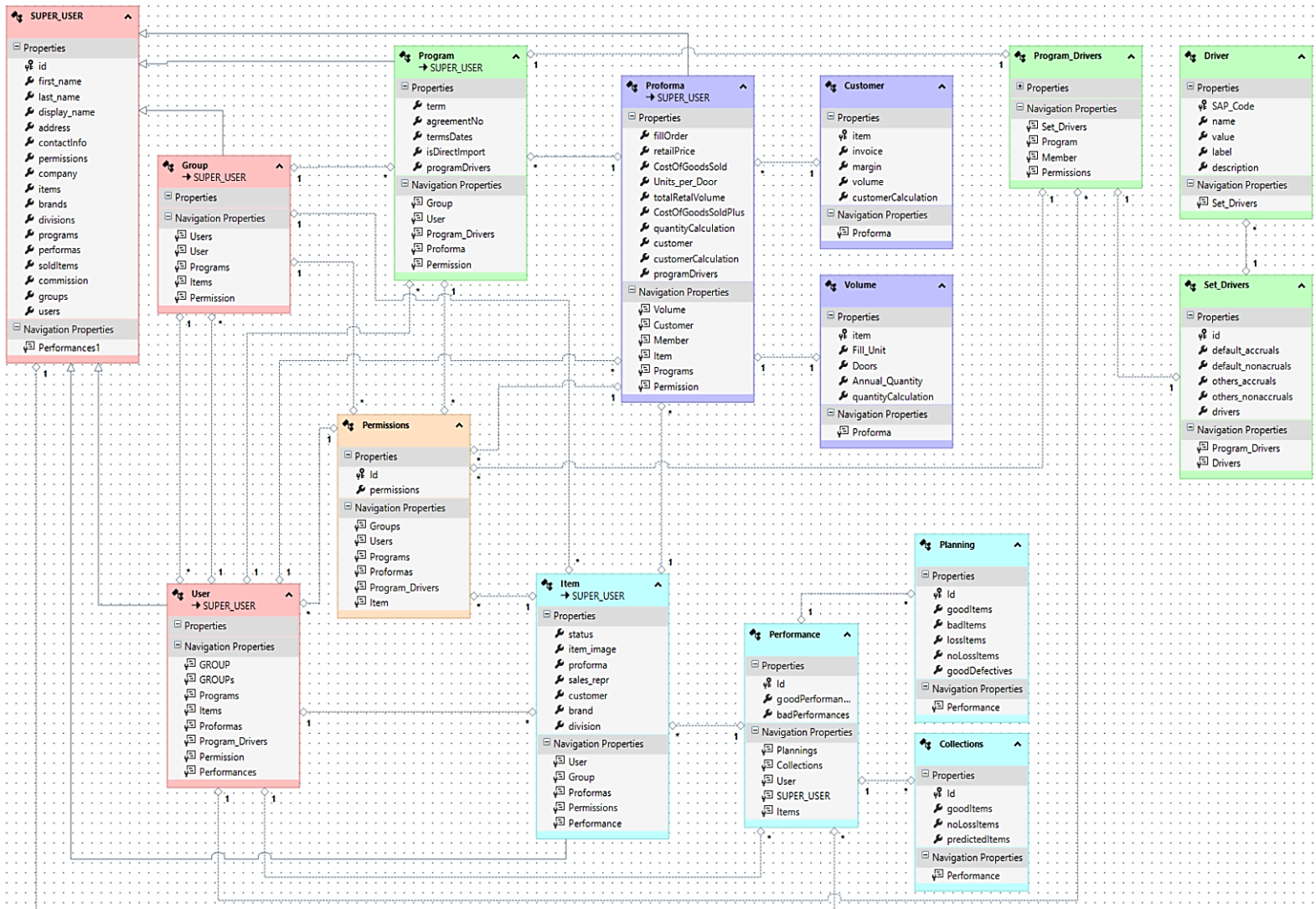








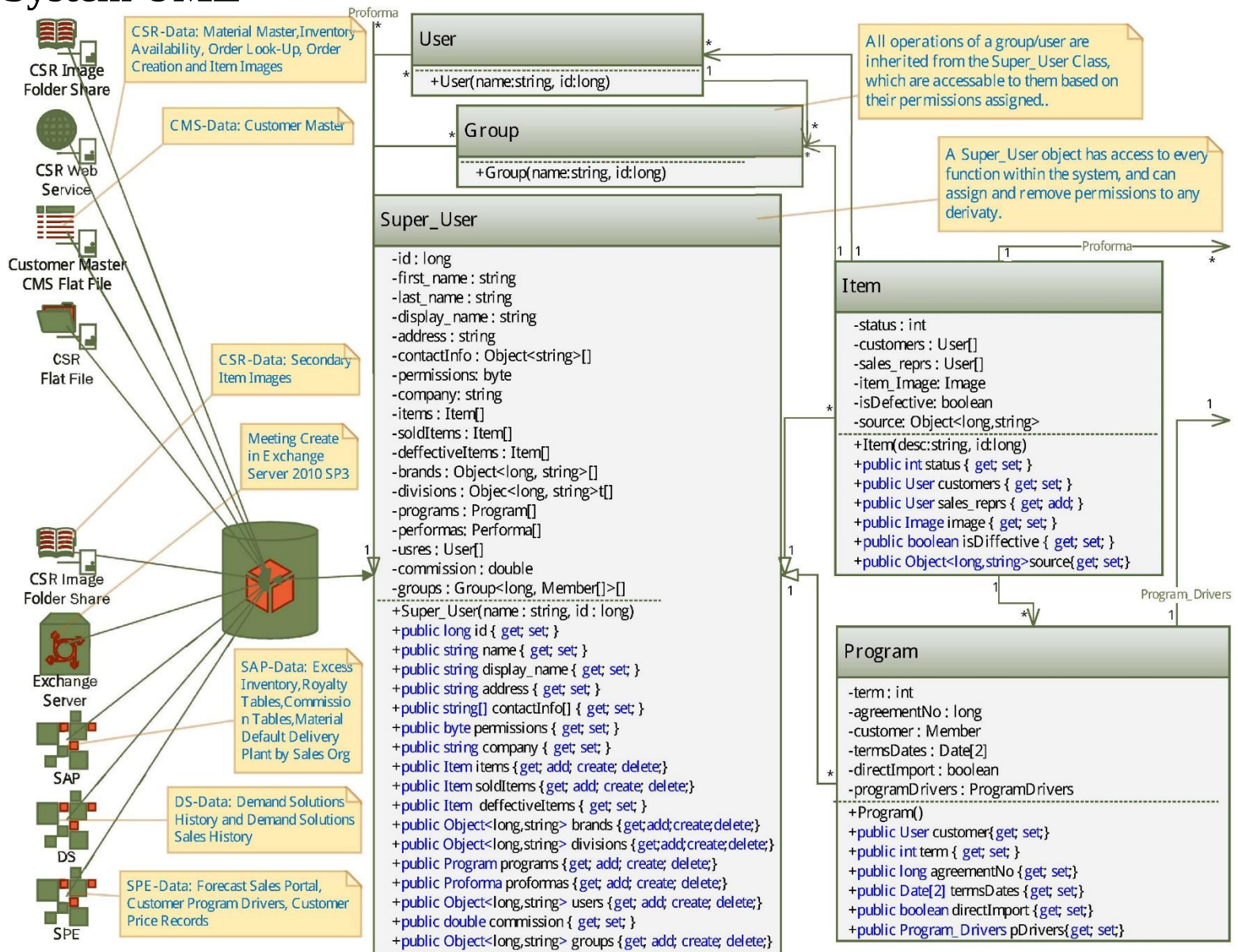
# Database Diagram – Color Coded



|                                                      |                                                                                                                                                                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Super_User, Group, User</b>                       | Objects utilized to reference a company, departments, a sales person, a customer, among other users.                                                                                                                        |
| <b>Program, Program_Drivers, Set_Drivers, Driver</b> | Objects utilized for pricing to a particular customer based on their agreement and item purchased.                                                                                                                          |
| <b>Proforma, Volume, Customer</b>                    | Objects utilized to provide a customer with a projection of the cost, price and profits generated by an item.                                                                                                               |
| <b>Item, Performance, Planning, Collections</b>      | Objects utilized to contain the information of a product(item), track its performance, plan to improve a product performance, and predict items that customers are more than likely to buy based on their purchase history. |
| <b>Permissions</b>                                   | Object utilized to privatize all functions in the Super_User and provide users with permissions to access all, or some, of these functions.                                                                                 |



# System UML



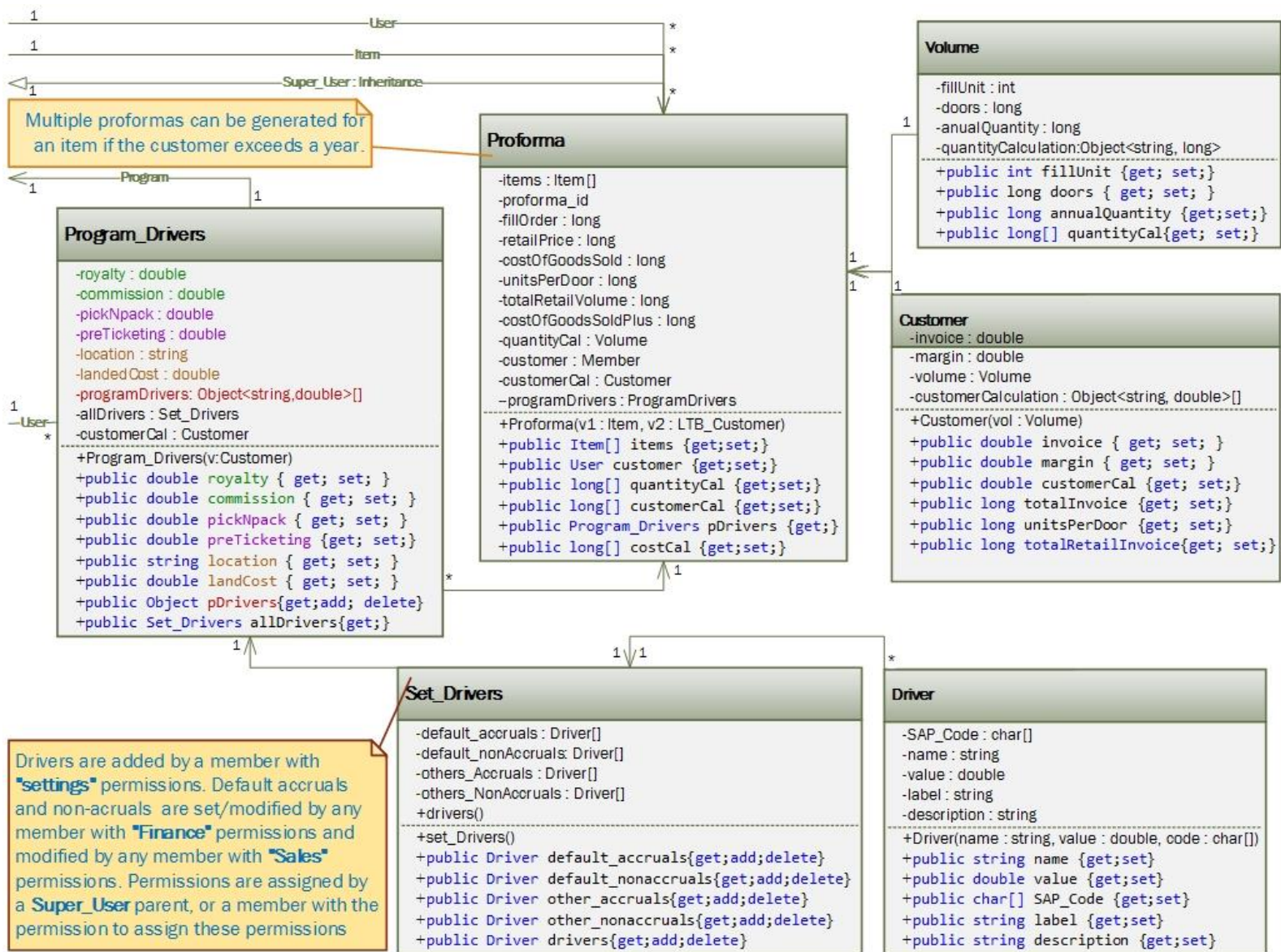
**Super\_User:** Main class that contains most and main functions of the system. These functions address every action that takes place in all groups, especially those in the finance, sales and customers. It is also responsible to save, update and access information, provided by the user, from/to the controllers/view.

**Groups:** Class used to relate a list of users to a unique user id that represents a unique group. This group inherits all functions in the Super\_User, which will be accessible according to the permissions set by a SUPER USER or a user with Employer permissions. Both users have permissions to create new users and, or, groups and provide these with permissions.

**User:** Class used to relate an employee, or customer, to a unique user id. The group class id will determine what type of user this is, such as finance person, sales person, customer, among others. The permissions of this user will also be set by a super user or a user with Employer permissions.

**Program:** Class used to set specific values that will drive the pricing of an item (product) for a specific customer. These drivers can only be set by a user with a finance group id.

**Item:** Class used to contain all information of an item, or product, and keep track of its status. Every user in this model is directly, or indirectly, related to an item. The selling process of an item involves a customer, sales person, division administrator and finance person in order to take place. The information of these, along with the properties of the item, are tracked and constantly updated.



**Program\_Drivers:** This class works along with the Program class to take care of the main drivers as well as accruals and non-accruals default drivers. This class can only be accessed by a user with Finance permissions.

**Set\_Drivers:** This class works along with the Program\_Drivers class to set the default accrual and non-accrual drivers of a specific customer. Same as Program\_Drivers a person with Finance permissions can only set these drivers.

**Driver:** This class is used to represent a unique driver for a specific customer. A list of this class form a list of values that drive the pricing of an item for a specific customer. Since this class is directly related to the Program, Program\_Drivers, and Set\_Drivers classes, these can only be set by a user with Finance permissions.

**Proforma:** A class used to calculate the cost of an item, its selling price, and the profit generated by this. This class makes use of the Program and User(customer) and Item objects to make the necessary calculations. This class can only be accessed by a user with Sales permissions, accepted by a user with higher Sales permissions, and validated by a user with Finance permissions.

**Volume:** This class is used to calculate the quantity, orders, among other properties of an item annually.

**Customer:** This class is used to calculate the annual invoice price, retail price and margin of a customer.





**Sales Full Permissions:** In this case, only a division administrator will have Full Permissions to a Proforma. This means that the administrator of a division will be able to accept and reject proformas of items in that division. Full Permissions could also include Employer Full Permissions.

**Sales Basic Permissions:** All sales people will have basic permissions, which can easily be modified by a SUPER USER or a user with the right permissions. These permissions will allow a sales person to create a proforma to sell an item.

**Finance Permissions:**

Finance permissions are those that an employee has to create a program and set values that will drive the pricing of an item sold by a sales person. Similarly, Finance has Full and Basic permissions, where Full permissions provided Employees with Employer Full Permissions and Basic permission do no.

**Finance Full Permissions:** Only finance managers will have full permissions to add, edit or remove an employee.

**Finance Basic Permissions:** All finance employees, even managers, have Basic permissions to allow them create, edit, and delete programs from the profile of a customer, and to validate proformas.

**Program:**

Programs are created by users with Finance permissions to set the global and default values that will drive the pricing of items for customers. A customer can have multiple programs since this varies based on the divisions that the items are from. Therefore, a user with finance permissions can create a program with a division and another program with another division generating in this way different pricing drivers for the same customer.

**Add/Set Drivers:** It is yet to determine what group does the user that add drivers belong to. Nevertheless, a SUPER USER or user with Employer permissions is in charge of providing the right user access to this function to add drivers.

**Item:**

An item object is created for every product that a company has, and they can be accessed by users with permissions. There are two types of permissions for an item, these are the Full permissions and Basic permissions.

**Item Full Permissions:** Only Sales people will have access to view and edit items properties as well as their status.

**Item Basic Permissions:** Every employee will be able to have basic permissions for items, which allow them to create and view items. All USERS, including customers, can view all items unless specified otherwise.

**Proforma:**

A proforma is created to provide customers with a projection of the calculated cost and price of an item as well as the customer and the company margins. Permissions for a proformas can also be Full and Basic.

**Proforma Full Permissions:** Only an administrator of a division will have full permissions of a Proforma, which allow the administrator to accept or reject the items pertaining to their division.

**Proforma Basic Permissions:** All Sales People will inherit Basic permissions to create, edit and delete a proforma.



## Permissions

|            |               |
|------------|---------------|
| <b>V</b> = | View          |
| <b>C</b> = | Create & View |
| <b>E</b> = | Edit & View   |
| <b>D</b> = | Delete & View |

- Functions: byte [] →

| V | C | E | D |
|---|---|---|---|
| 0 | 1 | 2 | 4 |

| Functions   |            | V | C | E | D |
|-------------|------------|---|---|---|---|
| User        |            |   |   |   |   |
| Group       |            |   |   |   |   |
| Program     |            |   |   |   |   |
| Proforma    | Validation |   |   |   |   |
| Item        | Status     |   |   |   |   |
| Planning    |            |   |   |   |   |
| Collection  |            |   |   |   |   |
| Performance |            |   |   |   |   |

## Permissions Table

|      |       |         | Proforma |            | Item |        |          |            |             |       |
|------|-------|---------|----------|------------|------|--------|----------|------------|-------------|-------|
| User | Group | Program | Proforma | Validation | Item | Status | Planning | Collection | Performance | Total |
| 7    | 7     | 7       | 7        | 7          | 7    | 7      | 7        | 7          | 7           | 70    |
| 6    | 6     | 6       | 6        | 6          | 6    | 6      | 6        | 6          | 6           | 60    |
| 5    | 5     | 5       | 5        | 5          | 5    | 5      | 5        | 5          | 5           | 50    |
| 4    | 4     | 4       | 4        | 4          | 4    | 4      | 4        | 4          | 4           | 40    |
| 3    | 3     | 3       | 3        | 3          | 3    | 3      | 3        | 3          | 3           | 30    |
| 2    | 2     | 2       | 2        | 2          | 2    | 2      | 2        | 2          | 2           | 20    |
| 1    | 1     | 1       | 1        | 1          | 1    | 1      | 1        | 1          | 1           | 10    |
| 0    | 0     | 0       | 0        | 0          | 0    | 0      | 0        | 0          | 0           | 0     |

## User Validation – Database Table Sample

| User Type     |      | Employer |       | Finance | Sales |      |          |            |             | Item |        |       |
|---------------|------|----------|-------|---------|-------|------|----------|------------|-------------|------|--------|-------|
|               | Id   | User     | Group | Program | Item  | Item | Planning | Collection | Performance | Item | Status | Total |
| Super User    | 0001 | 7        | 7     | 7       | 7     | 7    | 7        | 7          | 7           | 7    | 7      | 70    |
|               |      | 14       |       | 14      |       |      |          | 28         |             | 14   |        |       |
| Sales Admin   | 0002 | 7        | 2     | 0       | 1     | 7    | 3        | 3          | 0           | 3    | 2      | 28    |
|               |      | 9        |       | 1       |       |      |          | 13         |             | 5    |        |       |
| Sales Agent   | 0003 | 0        | 0     | 0       | 0     | 7    | 3        | 3          | 0           | 3    | 2      | 18    |
|               |      | 0        |       | 0       |       |      |          | 13         |             | 5    |        |       |
| Finance Agent | 0004 | 0        | 0     | 7       | 6     | 0    | 0        | 0          | 0           | 1    | 0      | 14    |
|               |      | 0        |       | 13      |       |      |          | 0          |             | 1    |        |       |
| Finance Admin | 0005 | 7        | 2     | 7       | 6     | 0    | 0        | 0          | 0           | 1    | 0      | 23    |
|               |      | 9        |       | 13      |       |      |          | 0          |             | 1    |        |       |
| Customer      | 0006 | 2        | 0     | 0       | 0     | 0    | 0        | 0          | 0           | 0    | 2      | 4     |
|               |      | 0        |       | 0       |       |      |          | 0          |             | 1    |        |       |

## Permissions Combination

| 0 | 1 | 2 | 4 |
|---|---|---|---|
| 0 | 0 | 1 | 2 |
| 1 | 1 | 2 | 3 |
| 2 | 2 | 3 | 4 |

|   |                |
|---|----------------|
| 0 | → View         |
| 1 | → Create, View |
| 2 | → Edit, View   |
| 4 | → Delete, View |

|   |                                        |
|---|----------------------------------------|
| 3 | → Reject, Accept, Validate             |
| 5 | → Reject, Accept, Invalidate           |
| 6 | → Validate, Invalidate                 |
| 7 | → Reject, Accept, Validate, Invalidate |

User Permissions Tables

| Groups Table |         |                 |
|--------------|---------|-----------------|
|              | User Id | Groups          |
| -1           | 0001    | 10XX (Sales)    |
| -2           |         | 11XX (Finance)  |
| -3           |         | 12XX (Employer) |
| -4           |         | 13XX (Item)     |

| Item Table |        |      |        |       |
|------------|--------|------|--------|-------|
| GroupId    | UserId | Item | Status | Total |
| 13XX       | 0001   | 7    | 7      | 14    |

| Employer Table |        |      |       |       |
|----------------|--------|------|-------|-------|
| GroupId        | UserId | User | Group | Total |
| 12XX           | 0001   | 7    | 7     | 14    |

| Finance Table |        |         |                     |       |
|---------------|--------|---------|---------------------|-------|
| GroupId       | UserId | Program | Proforma Validation | Total |
| 11XX          | 0001   | 7       | 7                   | 14    |

| Sales Table |        |          |          |             |             |       |
|-------------|--------|----------|----------|-------------|-------------|-------|
| GroupId     | UserId | Proforma | Planning | Collections | Performance | Total |
| 10XX        | 0001   | 7        | 7        | 7           | 7           | 28    |

Ways to Implement:

- 1) Through the Controller
- 2) PHP -> Front-End

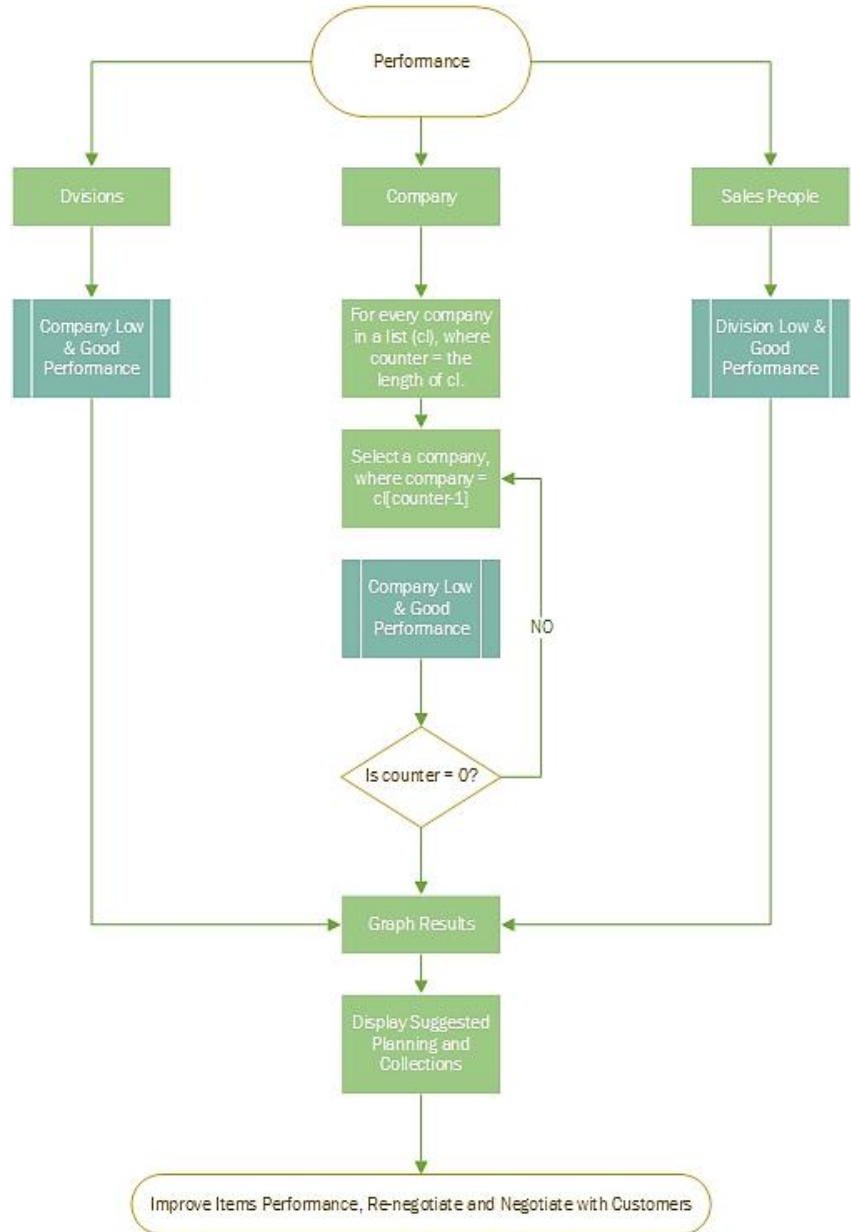
## LTB-Sales Performance

The performance of a company will be measured based on the performance of its divisions, which will be measure based on the sales people performance, which will be measured based on the items performance. In other words, the performance of the company will be indirectly related to the performance of its items (products). The performance can also be directly calculated if a straight analysis of all sold items is chosen.

**Company Level:** The performance of companies can be calculated by calculating the performance of the divisions of both companies in a recursive manner.

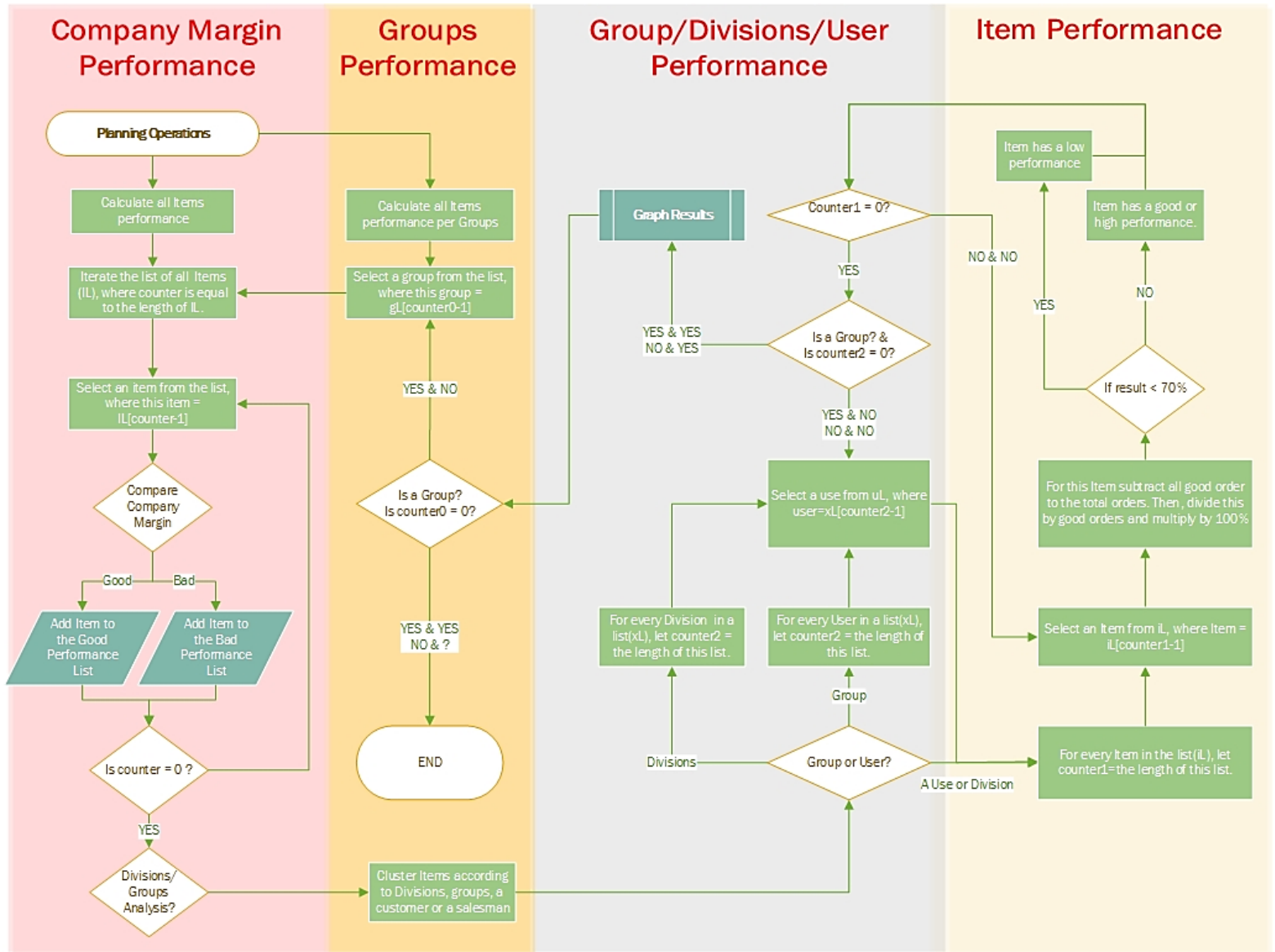
**Divisions Level:** The performance of all divisions can be calculated by recursively iterating through a division's list of Sales person.

**Sales People Level:** The performance of all sales people can be calculated by iterating through every sales person's list of sold items to calculate the performance of those items. First, the performance of the items can be obtained by clustering out the items that are below the expected margin, based on the quantity and value that the company expects. Second, clustering out the items that produced high cost because of being defectives, which should not affect the Sales person performance but the Company. Once the total number of sold items with a bad performance is found, this is subtracted to the number of items with a good performance and divided by the total of good sold items. The resulting number could be used to measure the performance of the sales person based on the good sales achieved. Consecutively, the performance of all Sales people will determine the performance of all Divisions and finally the performance of the Company, which will be directly affected by the high cost of returned and defective items.





# Company Margin Performance



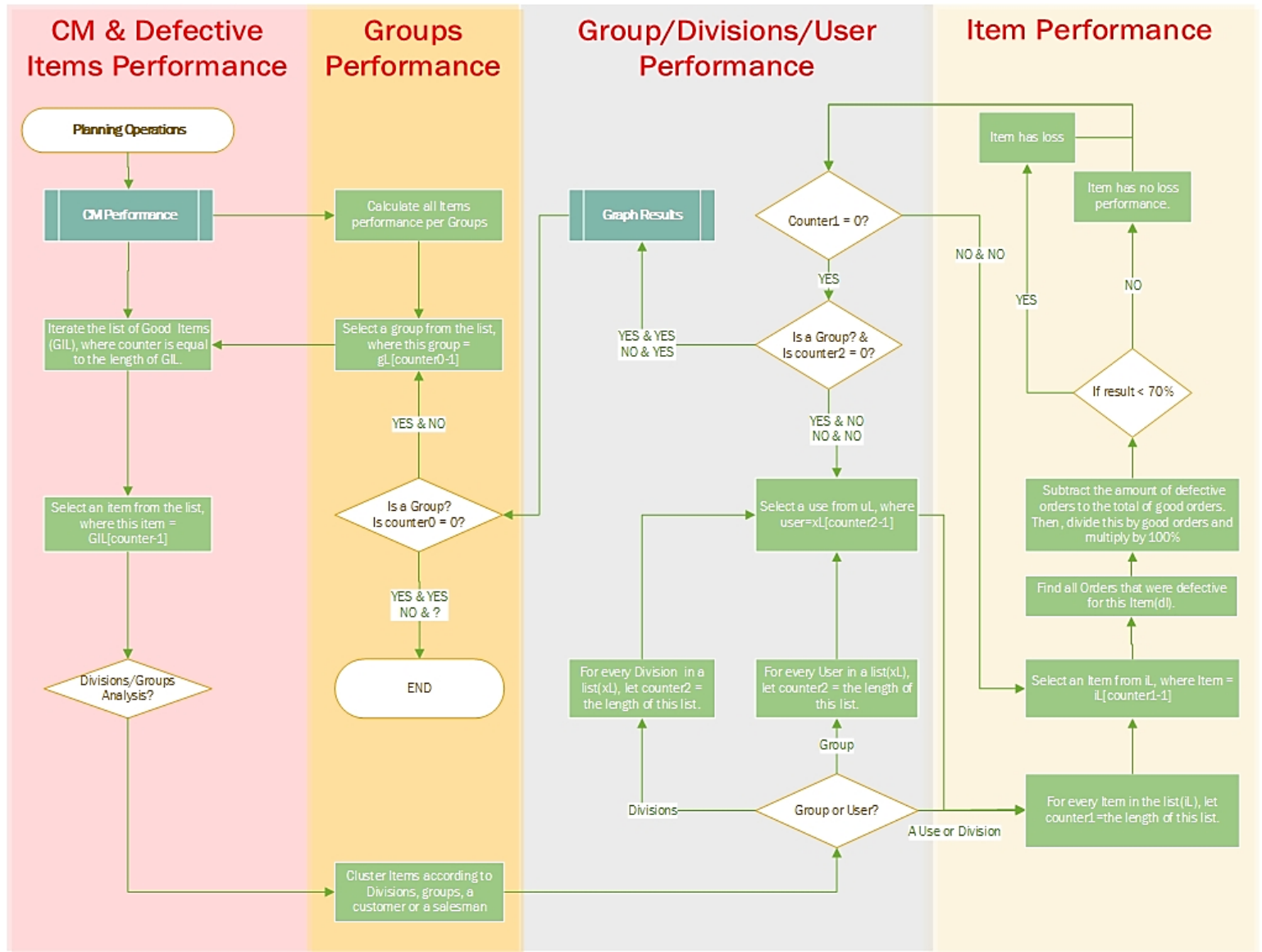
## Performances:

Company: A company can directly calculate the performance of all the items sold during certain period, or it can indirectly calculate its performance by calculating the performance of all divisions. Either way, the performance calculation will be based on the number of orders of an item with a good margin.

Groups: The performance of list of groups reclusively implements the group performance analysis.

Division/Group/User: Calculating the performance of a division, group and user is similar because they all have a list of items. The performance of an item is calculated by subtracting the item's number of orders, with a good margin, to the total number of orders of the item. Then, dividing it by number of good orders of that item.

# Company Margin and Defective Item Performance



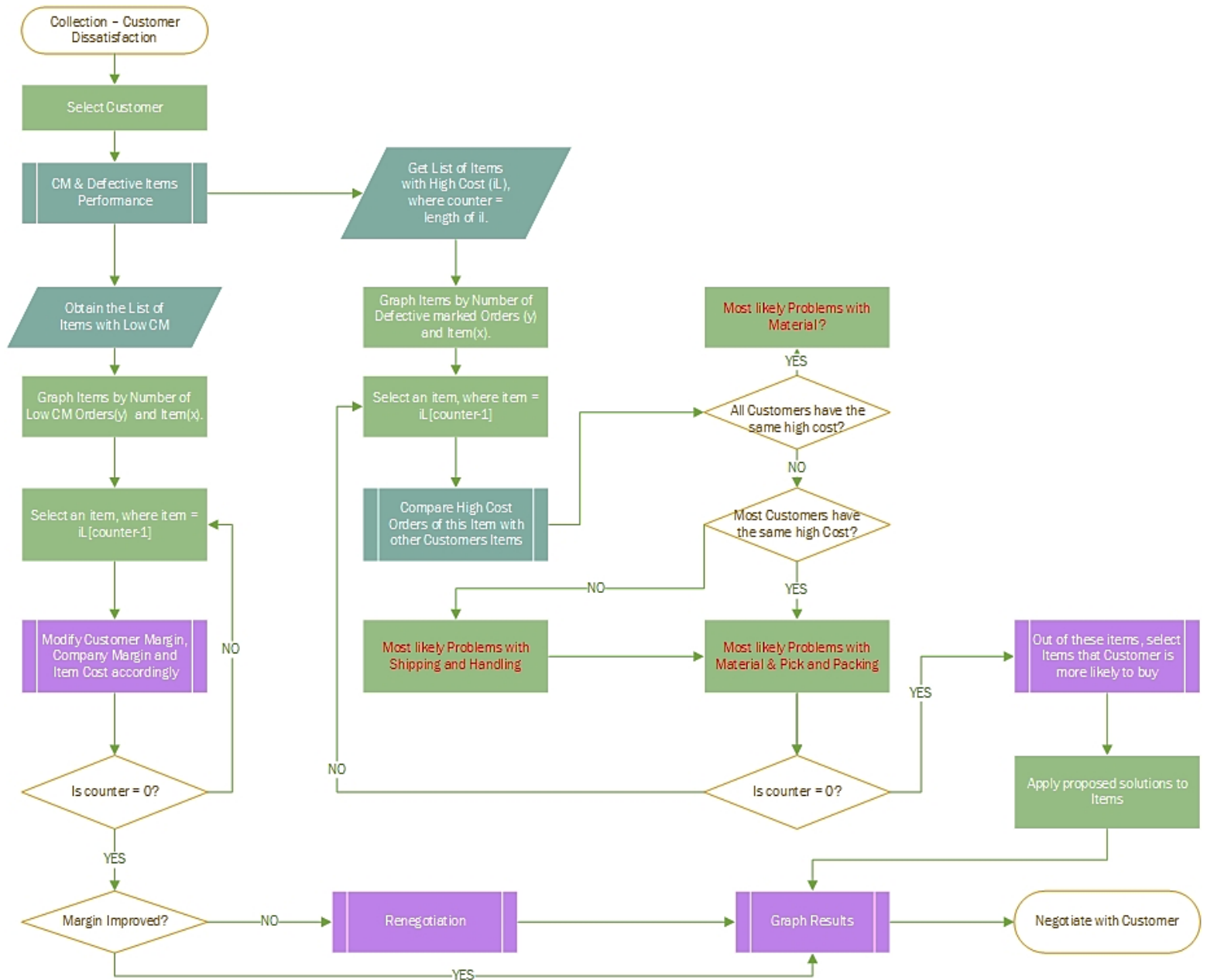
## Company Margin and Defective Item Performance

Company: In this performance analysis the items with a good margin are being realized by determining the number of orders of this items that were defective. As in the first performance analysis, the performance of the company can be analyzed indirectly by obtaining the performance of all divisions, and directly by directly obtaining the performance of all sold items.

Groups: The performance of list of groups reclusively implements the group performance analysis.

Division/Group/User: Calculating the performance of a division, group and user is similar because they all have a list of items. The performance of an item is calculated by subtracting the item's number of orders, with no defects, to the total number of orders of the item. Then, dividing it by number of good orders of that item.

# Planning and Collections



*\*High Cost: high number of orders in which items were defective*

## Obtain Items with Low Margin and High Cost of a particular customer:

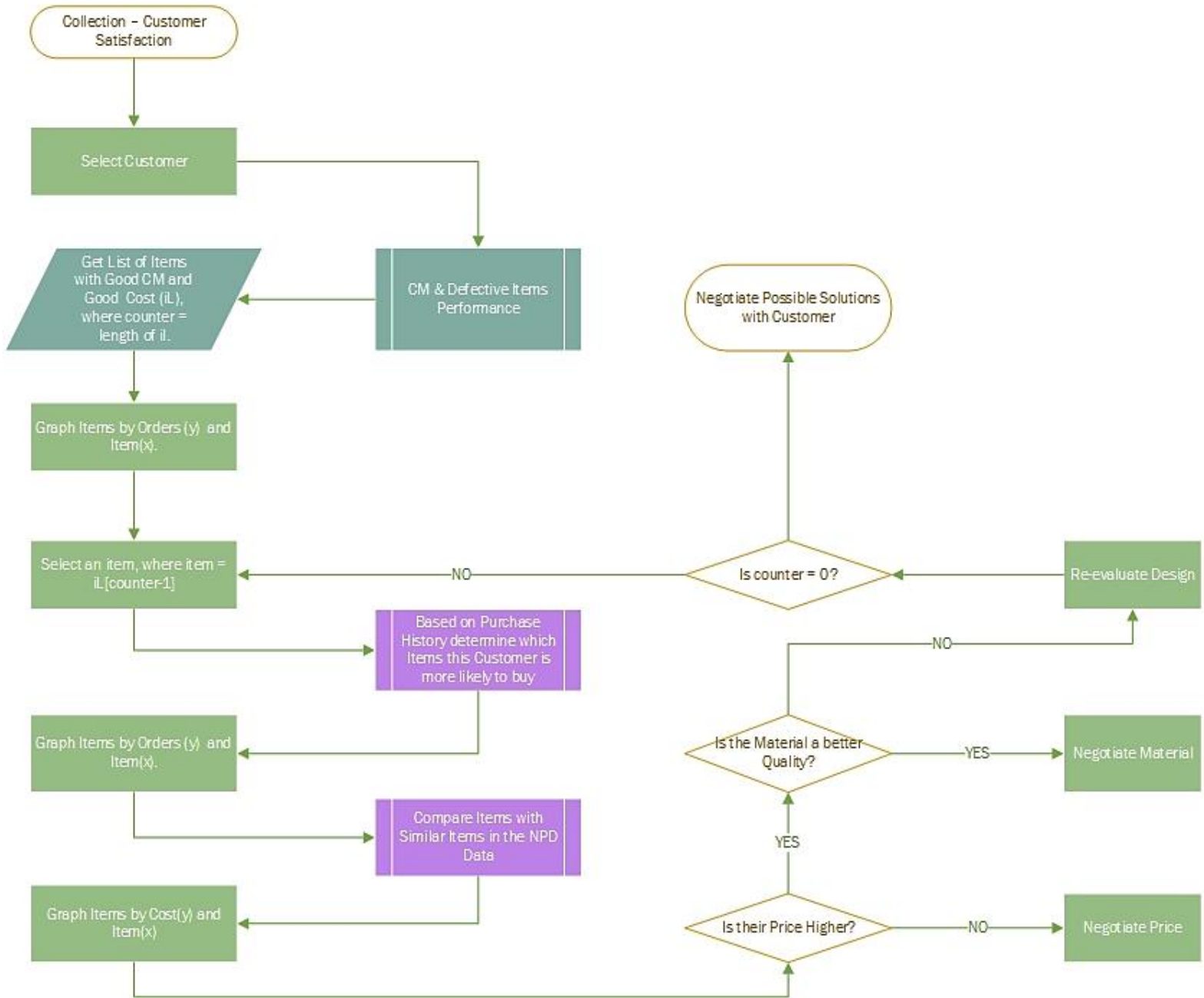
Using the performance function, the system can obtain the items with a bad, or low, margin as well as those with a high cost. After obtaining these items from a specific customer, the planning-collections function can be used to improve the items performance and negotiate with the customer.

**Modify customer margin, company margin and, or, item cost accordingly:** Once the planning-collections function is implemented the Sales person will have to manually make modifications to the Customer Margin, Company Margin and, or, Item cost accordingly. These modifications must be done, if possible, taking into consideration that the customer and company margins should not be affected. If this does not improve the Item's performance, then re-negotiation must be taken into consideration.

**Compare High Cost with other customers that have purchased the same item in this company:** For items with High Cost, the system will take a different approach. These items will be compared to customers that have purchased the same items. In this manner, the system can analyze if the number of defective items per orders could be related to the material, shipping and or pick and packing.



# Planning & Collections



**Obtain all Items, with a Good Performance, from a specific Customer:** Using the performance function, the system can obtain the items with a good performance. After obtaining these items from a specific customer, the planning-collections function can be used to generate a list of items that the customer is more than likely to buy, and compare these to similar items in other companies to understand why they haven't bought them yet.

**Items Prediction based on Purchase History:** Based on the purchase history of the items that a specific customer has bought, the system could predict which items this customer is more likely to buy. To have a better prediction of these items is best to use those that the customer has had a good experience with. In this manner, the customer can relate a positive connection between items.

**Compare Items with other similar Items from Different Companies:** After predicting the items that a customer might like, the question to address is why haven't they bought these items? To answer this question, the system could compare the items in the predicted-items list with other similar items in other companies. They may be purchasing these items from other companies because the material they use is cheaper and, or, the price of this item is cheaper.